Vector addition and decomposition

LEARNING TARGET · S2 I can join arrows tip to tail and split an arrow into parts using sine and cosine.

A · KEY VOCABULARY

Vector	A quantity with size and direction, drawn as an arrow.
Tip-to-tail	Start each arrow where the last one ends. Do not turn or stretch it.
Component	The east–west or north–south part of a vector.
Resultant	The overall arrow from the first start to the final finish.

B · KEY EQUATIONS + SIGNS

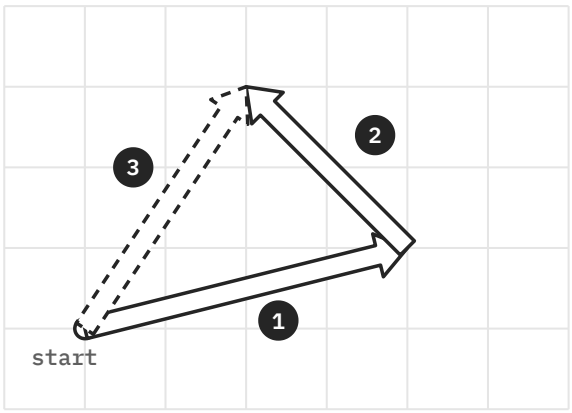
$$R_x = A_x + B_x \cdot R_y = A_y + B_y$$
$$|R| = \sqrt{(R_x^2 + R_y^2)}$$
$$\theta = \tan^{-1}(R_y \div R_x)$$

parts: V cos θ (next to θ) · V sin θ (opposite)

- \+ east, north
- – west, south

C · COLOR IT · TIP TO TAIL

Each square = 1 m. No trig needed.



A: 4 E, 1 N

B: 2 W, 2 N

sum: 2 E, 3 N

E–W: 4 – 2 = 2 E

N–S: 1 + 2 = 3 N

☐ **1 · BLUE** Arrow A

☐ **2 · BLUE** Arrow B

☐ **3 · BLUE** Resultant (dashed)

D · WORKED EXAMPLE

A 20 N pull points 30° above the +x axis. Split it into its x and y parts.

GIVEN	EQUATION	WORK	ANSWER
F = 20 N θ = 30° from +x	$F_x = F \cos \theta$ $F_y = F \sin \theta$	20 cos 30° 20 sin 30°	**17.3 N, 10.0 N** parts of one force, not two new forces

E · THE 2026-XJ THREAD

DAY 6 OF 22 · NEW TOOL: VECTOR ADDITION

XJ's motion has parts in more than one direction. Adding them tip to tail gives its true path; splitting them shows the part that actually closes on Earth.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else